CSE 534: Fundamentals of Computer Networks
Project Progress Report

LLM-Augmented Reasoning for IoT Network Traffic Classification

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1. Overview and Recap

This report summarizes the progress made on our course project, which aims to apply LLM-based reasoning agents alongside conventional ML classifiers for IoT network traffic classification. We are using the CIC-BCCC-NRC TabularIoTAttack-2024 dataset, which contains over 620,000 labeled network flow records extracted via CICFlowMeter across four traffic classes: benign, DDoS UDP Flood, DoS TCP Flood, and Reconnaissance Port Scan. Since the proposal, we have completed the first phase of our planned work—exploratory data analysis (EDA)—and have refined our project architecture into a two-track system incorporating both zero-shot LLM classification and ML-classifier-driven explainability.

2. Work Completed: Exploratory Data Analysis

We conducted a thorough EDA on the dataset, covering structural inspection, class distribution analysis, univariate and bivariate feature profiling, correlation analysis, and dimensionality reduction. The key findings are summarized below.

2.1 Dataset Structure. The combined dataset comprises 620,718 flow records across 85 columns (84 CICFlowMeter features plus the traffic label). No missing values were found across any feature column. After dropping non-analytical identifier columns (Flow ID, Source IP, Destination IP, Timestamp), 81 numeric features remained for analysis.

2.2 Class Distribution. The dataset exhibits significant class imbalance. Reconnaissance traffic dominates with 485,522 records (78.2%), followed by DoS TCP Flood at 100,000 (16.1%), Benign at 32,620 (5.3%), and DDoS UDP Flood at 2,576 (0.4%). This imbalance is an important consideration for classifier training and evaluation, and we plan to use stratified sampling, class-weighted loss functions, and macro-averaged metrics to handle it.

Traffic Class	Record Count	Percentage
RECON	485,522	78.2%
DOS_TCP	100,000	16.1%
BENIGN	32,620	5.3%
DDOS_UDP	2,576	0.4%

2.3 Feature Distributions and Boxplot Analysis. We profiled five core traffic features: Flow Duration, Total Fwd Packets, Total Backward Packets, Flow Bytes/s, and Flow Packets/s. Boxplot comparisons across classes revealed distinct behavioral signatures. DDoS UDP Flood traffic exhibited extremely high packet and byte rates with heavy-tailed distributions and significant outliers, consistent with volumetric flooding behavior. DoS TCP Flood showed elevated forward packet counts and moderate byte throughput. Reconnaissance traffic was characterized by shorter flow durations and repetitive probing patterns. Benign traffic showed balanced bidirectional exchange with moderate, stable rates.

2.4 Correlation Analysis. A correlation heatmap across the selected features showed a strong positive relationship between Flow Bytes/s and Flow Packets/s, consistent with flood-style attacks that scale both metrics simultaneously. Flow Duration exhibited weak to moderate correlation with rate-based features, indicating that short-lived probes and sustained floods represent distinct attack mechanisms that contribute different discriminative information to classifiers.

2.5 SYN Flag Analysis. SYN Flag Count analysis revealed consistent low-but-present SYN activity in reconnaissance traffic, reflecting systematic TCP connection probing. DoS TCP Flood exhibited elevated SYN counts tied to handshake abuse. DDoS UDP traffic showed minimal SYN activity, expected for UDP-based floods. This transport-layer feature provides important discriminative signal across attack types.

2.6 Dimensionality Reduction. PCA projection onto two principal components showed partial class separation, indicating that some attack types differ along broad linear trends in duration, packet count, and rate. t-SNE visualization revealed tighter, more distinct clusters, suggesting that non-linear feature interactions carry significant discriminative information. Reconnaissance traffic formed especially compact clusters, consistent with its repetitive probe signature. These results confirm that both linear and non-linear classifiers are viable, but non-linear approaches (or reasoning-based methods like LLMs) may capture richer structure.

3. Revised Project Architecture

Based on EDA findings and discussions with the instructor, we have refined our project into a **two-track architecture** that operates in parallel and then merges:

Track A – Zero-Shot LLM Agent. An LLM agent receives serialized flow records and performs classification using only its pre-trained knowledge of network protocols, attack signatures, and traffic semantics—without any task-specific training data. The agent uses chain-of-thought reasoning to analyze feature values (e.g., abnormally high Flow Packets/s, zero backward packets, elevated SYN flags) and produces a classification label along with a natural-language explanation of why a given flow is, for example, a DDoS attack versus a DoS attack versus benign traffic. This track tests whether LLMs can reason about network security from first principles.

Track B – ML Classifier + LLM Explainer. We train conventional ML classifiers (Random Forest, XGBoost) on the labeled dataset to achieve high-accuracy predictions. The trained classifier's output, along with the input features and feature importances (e.g., SHAP values), are then passed to an LLM that generates a human-readable explanation of the prediction. For instance, if the classifier labels a flow as DDOS_UDP, the LLM explains: "This flow was classified as a DDoS UDP Flood because it exhibits an exceptionally high packet rate (Flow Packets/s = 245,000) with zero backward packets and minimal flow duration, consistent with a unidirectional volumetric flood."

Merge. We will compare Track A (zero-shot) and Track B (trained + explained) across accuracy, F1-score, and explanation quality. The final system can potentially combine both: the ML classifier for fast, accurate prediction and the LLM for interpretable justification, with the zero-shot LLM serving as a second opinion for ambiguous or borderline cases.

4. Remaining Work and Timeline

Week	Task	Deliverable
Week 9–10	Train RF, XGBoost; hyperparameter tuning; handle class imbalance	Baseline classifier metrics
Week 10–11	Design LLM agent prompts; zero-shot classification experiments	Zero-shot accuracy report
Week 11–12	SHAP feature importance; LLM explainer integration with ML classifier	Explained predictions pipeline
Week 12–13	Merge tracks; comparative evaluation; ablation studies	Unified evaluation report
Week 14	Final report writing and presentation preparation	Final report + slides

5. Key Takeaways from EDA

The EDA phase has yielded several insights that directly inform our subsequent modeling decisions:

(1) Strong class-specific signatures exist: DDoS traffic is identifiable by extreme packet/byte rates; reconnaissance by short durations and SYN probing; DoS TCP by elevated forward packets. These clear behavioral patterns suggest that both ML classifiers and LLM reasoning agents have sufficient signal for accurate classification.

(2) Class imbalance requires careful handling: With DDOS_UDP comprising only 0.4% of the data, stratified evaluation and class-weighted training are essential. For the LLM track, few-shot exemplars must be balanced across classes.

(3) Non-linear separability: The contrast between PCA (partial overlap) and t-SNE (distinct clusters) confirms that non-linear feature interactions are important. This motivates both tree-based classifiers and the use of LLM reasoning, which can consider feature combinations holistically.

(4) Explainability is feasible: The semantic meaning of CICFlowMeter features (packet rates, flag counts, durations) maps naturally to network security concepts that an LLM can articulate, making the explanation task well-suited to language model capabilities.